Hicham Randrianarivo

Principal Research Scientist

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Profile

Principal Research Scientist specializing in LLM-based retrieval and generative retrieval. Develops methods for document/semantic ID generation, hallucination detection, and rigorous evaluation for retrieval systems at scale.

Selected Impact

Retrieval at scale	Developed and operated Qwant image search at 400 QPS with a 15B-document index and 1M updates/day; addressed safety complaints from upstream content.
Model quality	Improved Huawei keyword extraction and video classification accuracy; shipped two models: +22 % on video classification and +15 % on keyword extraction.
Generative retrieval	Fine-tune small LLMs (<8B parameters) to serve as retrieval indexes that generate document/semantic IDs for document retrieval.
LLM reliability	Develop hallucination detection methods and evaluation datasets for LLM retrieval pipelines [8].
LLM pretraining	Contribute to continual pretraining (CPT) of the ArGiMi Mistral 24B model.

Research Focus & Skills

LLM Modeling	Generative retrieval methodology: fine-tuning models to generate document/semantic IDs and designing ID spaces optimized for generatability.
LLM Evaluation	Hallucination detection, benchmark construction, and evaluation protocols for LLM-based retrieval.
Retrieval Systems	ANN search, indexing pipelines, and ranking models for large-scale retrieval.
Computer Vision	Image/video retrieval, multimodal representation learning, and semantic segmentation.

Research Experience

Oct 2024 – Present **Principal Research Scientist**, *Artefact Research Center*, Paris
Research on ArGiMi, a BPI-funded joint program with Giskard, Mistral AI, INA, and BnF to build deployable LLM retrieval systems and evaluation assets for enterprise search (ArGiMi consortium).

- Contribute to continual pretraining (CPT) of the ArGiMi Mistral 24B model trained for the project.
- Define LLM-based retrieval assets with models under 8B parameters, balancing quality, latency, and cost.
- Develop generative retrieval methods, training models to index documents by generating document/semantic IDs via parametric indexing.
- Design a unified objective that optimizes retrieval and generation together, reducing pipeline misalignment and improving reliability.
- Develop evaluation datasets and protocols for business-oriented tasks to quantify hallucination and retrieval failures in LLM pipelines [8].
- Maintain the open-source `artefactual` repo for LLM generation evaluation (30 stars in the first year).

Oct 2024 – Present **Associate Researcher**, *CentraleSupélec (MICS Laboratory)*, Paris-Saclay

Aug 2022 – Mar 2024 **Senior Research Engineer**, *Huawei*, Amsterdam
Applied research for Petal Search, building multimodal retrieval models and production features.

- Improved keyword extraction and video classification accuracy; shipped two models: $+22\,\%$ video classification and $+15\,\%$ keyword extraction.
- Built multimodal LLM pipelines for video understanding, generating keywords and summaries from proprietary datasets.
- Led an AI code assistant project using CodeT5 and CodeLlama, including code search and retrieval capabilities.
- Prototyped multi-view 3D reconstruction for object retrieval and advised engineering teams on productionization.

Jun 2017 – Aug 2022 **Research Engineer**, *Qwant*, Paris
Research engineer on the image search engine, delivering retrieval systems at scale.

- Built the image search engine end-to-end; monitored users, index growth, and safety complaints from upstream content.
- Operated the image retrieval system at 400 QPS with a 15B-document index and 1M document updates/day.
- Architected a content-based image retrieval stack with scalable indexing and multimodal embeddings to improve relevance and efficiency [10, 11].
- Co-developed the SnapEarth satellite image search engine and built the EarthSignature and EarthSearch modules.
- Initiated and managed a joint research project with Graphcore to benchmark IPU inference for retrieval models (cf. [4]).

Sep 2012 – Dec 2016 **PhD Student**, *Onera, the French aerospace lab.*, Paris
PhD research on semantic interpretation of aerial images for large-scale retrieval and indexing [12].

- Developed statistical learning models for semantic segmentation using mixtures of object detectors to improve image understanding [14].
- Designed shape models and structural methods for superpixel classification to improve segmentation accuracy [13].

Mar 2012 – Sep 2012 **Research Intern**, *Laboratory of Computer Sciences, Paris 6 (LIP6)*, Paris
Researched large-scale image classification and retrieval.
- Developed kernel approximation methods enabling linear-time training in high-dimensional spaces.

Jul 2011 – Aug 2011 **Research Intern**, *ETIS (Information Processing and System Teams)*, Cergy-Pontoise
Evaluated machine learning methods for image retrieval, benchmarking performance and scalability.

Education

2012 – 2016 **PhD**, *Conservatoire National des Arts et Métiers*, Paris
PhD. in computer science. Thesis focused on statistical learning for semantic interpretation of aerial images, contributing to advanced image analysis methodologies relevant to information retrieval.

2011 – 2012 **Master of Science**, *University of Cergy-Pontoise*, Cergy-Pontoise
Master of Science in Computer Vision and Machine Learning. Dual degree with engineering program, emphasizing advanced algorithmic development crucial for retrieval systems.

2009 – 2012 **Engineering degree**, *ENSEA*, Cergy-Pontoise
Engineering degree with specialization in Applied Mathematics and Computer Vision, foundational for research in AI and image processing, particularly in information retrieval contexts.

Languages

French Native
English Proficient

Publications

[1] Nicolas Audebert, Alexandre Boulch, Hicham Randrianarivo, Bertrand Le Saux, Marin Ferecatu, Sébastien Lefevre, and Renaud Marlet. "Deep learning for urban remote sensing". In: *2017 Joint Urban Remote Sensing Event (JURSE)*. IEEE. 2017, pp. 1–4.

[2] Manuel Campos-Taberner, Adriana Romero-Soriano, Carlo Gatta, Gustau Camps-Valls, Adrien Lagrange, Bertrand Le Saux, Anne Beaupere, Alexandre Boulch, Adrien Chan-Hon-Tong, Stephane Herbin, et al. "Processing of extremely high-resolution Lidar and RGB data: outcome of the 2015 IEEE GRSS data fusion contest–part a: 2-D contest". In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 9.12 (2016), pp. 5547–5559.

[3] Chen Dang, Hicham Randrianarivo, Raphaël Fournier-S'Niehotta, and Nicolas Audebert. "Web Image Context Extraction with Graph Neural Networks and Sentence Embeddings on the DOM tree". In: *Joint European Conference on Machine Learning and Knowledge Discovery in Databases 3rd Workshop on Graph Embedding and Mining* (2021).

[4] Ilyes Kacher, Maxime Portaz, Hicham Randrianarivo, and Sylvain Peyronnet. *Graphcore C2 Card performance for image-based deep learning application: A Report*. 2020. arXiv: `2002.11670 [cs.CV]`.

[5] Adrien Lagrange, Bertrand Le Saux, Anne Beaupere, Alexandre Boulch, Adrien Chan-Hon-Tong, Stéphane Herbin, Hicham Randrianarivo, and Marin Ferecatu. "Benchmarking classification

of earth-observation data: From learning explicit features to convolutional networks". In: *2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE. 2015, pp. 4173–4176.

[6] Bertrand Le Saux and Hicham Randrianarivo. "Urban change detection in SAR images by interactive learning". In: *2013 IEEE International Geoscience and Remote Sensing Symposium-IGARSS*. IEEE. 2013, pp. 3990–3993.

[7] Bertrand Le Saux, Hicham Randrianarivo, and Nicolas Audebert. *Christchurch Aerial Semantic Dataset*. Dec. 2019. DOI: 10.5281/zenodo.3566005. URL: https://doi.org/10.5281/zenodo.3566005.

[8] Charles Moslonka, Hicham Randrianarivo, Arthur Garnier, and Emmanuel Malherbe. *Learned Hallucination Detection in Black-Box LLMs using Token-level Entropy Production Rate*. Accepted at ECIR 2026. 2025. arXiv: 2509.04492 [cs.CL]. URL: https://arxiv.org/abs/2509.04492.

[9] Adrien Nivaggioli and Hicham Randrianarivo. "Weakly supervised semantic segmentation of satellite images". In: *2019 Joint urban remote sensing event (JURSE)*. IEEE. 2019, pp. 1–4.

[10] Maxime Portaz, Adrien Nivaggioli, Hicham Randrianarivo, Ilyes Kacher, and Sylvain Peyronnet. "QISS: An Open Source Image Similarity Search Engine". In: *Advances in Information Retrieval*. Springer International Publishing, 2020, p. 486.

[11] Maxime Portaz, Hicham Randrianarivo, Adrien Nivaggioli, Estelle Maudet, Christophe Servan, and Sylvain Peyronnet. *Image search using multilingual texts: a cross-modal learning approach between image and text*. 2019. arXiv: 1903.11299 [cs.CV].

[12] Hicham Randrianarivo. "Apprentissage statistique de classes sémantiques pour l'interprétation d'images aériennes". PhD thesis. Conservatoire national des arts et metiers-CNAM, 2016.

[13] Hicham Randrianarivo, Bertrand Le Saux, Nicolas Audebert, Michel Crucianu, and Marin Ferecatu. "Structural classifiers for contextual semantic labeling of aerial images". In: 2016.

[14] Hicham Randrianarivo, Bertrand Le Saux, Michel Crucianu, and Marin Ferecatu. "Discriminatively-trained model mixture for object detection in aerial images". In: *Image Information Mining, Bucharest, Romania* (2015).

[15] Hicham Randrianarivo, Bertrand Le Saux, and Marin Ferecatu. "Multimodal classification with deformable part-based models for urban cartography". In: *2014 IEEE Geoscience and Remote Sensing Symposium*. IEEE. 2014, pp. 203–206.

[16] Hicham Randrianarivo, Bertrand Le Saux, and Marin Ferecatu. "Urban structure detection with deformable part-based models". In: *2013 IEEE International Geoscience and Remote Sensing Symposium-IGARSS*. IEEE. 2013, pp. 200–203.